Mini Project Design

Designing, Programming and Testing an LP Management System

13.04.2026 – 17.04.2026

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Introduction

Purpose

Through working on a small example, you will gain an understanding of the basic activities and deliverables involved in software development.

Goals

- Demonstrate an understanding of the open 3-layered architecture (user interface layer – application control layer – model layer) and be able to implement it.
- Design the distribution of responsibilities between controller classes and domain classes using interaction diagrams (UML communication diagrams).
- Develop design class diagram based on interaction diagrams, domain model etc.
- Implement use cases according to the design.
- Integrate individual parts of a system into a complete system.
- Implement container classes to manage the domain objects.
- Implement different types of structures between objects (e.g. association, aggregation).
- Test selected controller and container classes.
- Use version control.

How to Work

Work cooperatively in groups. You are responsible for organizing your work and dividing tasks among your group. When implementing the application, working in pairs is highly recommended. Active participation from all members is essential. If challenges arise, first attempt to resolve them within your group. If further assistance is needed, consult the lecturer.

The system development will follow an iterative process using the Unified Process (UP) methodology. You are currently in the Elaboration phase. During the Inception phase and the initial iteration of Elaboration, the use cases were identified and prioritized based on complexity and business value. In this iterative approach, development will start with the highest priority use case.

Project schedule and resources

Provisional Schedule for the Mini project

Day 1: Introduction to the case, exercises, and the code
Start the first iteration: Design, implementation, and test of “Lend LP copy”.

Day 2: Continue: Design, implementation, and test of the current use case. When (and if) you are done with current use case, you continue with the next use case from the prioritized list.
Review of design documents for the first use case.

Day 3: Continue: Design, implementation, and test of the current use case. When (and if) you are done with current use case you continue with the next use case from the prioritized list.

Day 4: Continue: Design, implementation, and test of the current use case. When (and if) you are done with current use case you continue with the next use case from the prioritized list.

Day 5: Complete: Design, implementation, and test of the current use case.
Complete: Documentation in the report.
Submit: Your work in Wiseflow by 14.00.
Begin: Preparation for evaluation.

A programming or system development lecturer will be available on select days as outlined in your Calendar.

Hand-out

- Description of the problem
- Diagrams
- Project Example with TUI
- Project Example with Generic TUI

Tools

BlueJ (code), UMLet (diagrams), Git, GitHub, GitHub Desktop, Word, Mendeley

Version Control

You must invite your lecturers to your Git repository in GitHub.

Documentation Requirements

- The report must not exceed 15 normal pages. Definition of one normal page: 2400 characters including white spaces and footnotes. Not including cover page, table of contents, reference list and appendices. The following diagrams and artefacts must be included in the report:
    - Fully dressed use case and SSD (system sequence diagram) for the non-trivial use cases.
    - Operation contracts for the relevant system operations.
    - Description of the architecture.
    - Interaction diagrams for the implemented use cases. Discuss your design decisions.
    - Design class diagram. Explain how it is deducted from the interaction diagrams. Remember to add all methods (getters and setters and UI methods).
    - Coding standard.
    - Code examples and discussion of these.
    - Descriptions of the unit tests that were implemented.
    - Group Collaboration Contract.
    - Evaluation of group work.
    - Remember references to all sources used (books, websites etc).

Furthermore, you need to meet the formal requirements found in the Institutional section in the Curriculum (https://www.ucn.dk/uddannelser/datamatiker/uddannelsens-indhold/studieordning-og-officielle-dokumenter).

Hand-in

- Day 5 by 14:00.
- Written documentation as described above in Documentation Requirements **as a Report** in form of a **PDF-file** including following information:
  - All diagrams (from *Documentation Requirements*)
  - Repository path (on title page)
  - Latest commit (on title page)
  - Number of commits (on title page)
- Zipped copy of the project code must be included in the Wiseflow hand-in
- **The report & the zipped project** are to be:
  - uploaded to Wiseflow.
- See "VejledningRapportSkrivning (2026).pdf"

Evaluation

The evaluation will take place on the **23rd of April 2026 – one group at a time.**
We will use at most **40 minutes** per group. The procedure will be as follows:
*The group will **present** their results using maximum **10 minutes** (remember your experience from last time and make a schedule in the group).*
Questions and feedback from the lecturers are given in the remaining time. Programming and System Development lecturers will be present.

Schedule

Hour	Group
08.30 – 09.10	1
09.10 – 09.50	2
09.50 – 10.00	Break
10.00 – 10.40	3
10.40 – 11.20	4
11.20 – 12.00	Break
12.00 – 12.40	5
12.40 – 13.20	6

Case Description and Development Tasks

An application to manage the lending of your LP records (also known as vinyl) to friends is to be developed. The application will store information about your friends and your LP collection, including multiple copies of the same LP.

Key features of the application include:

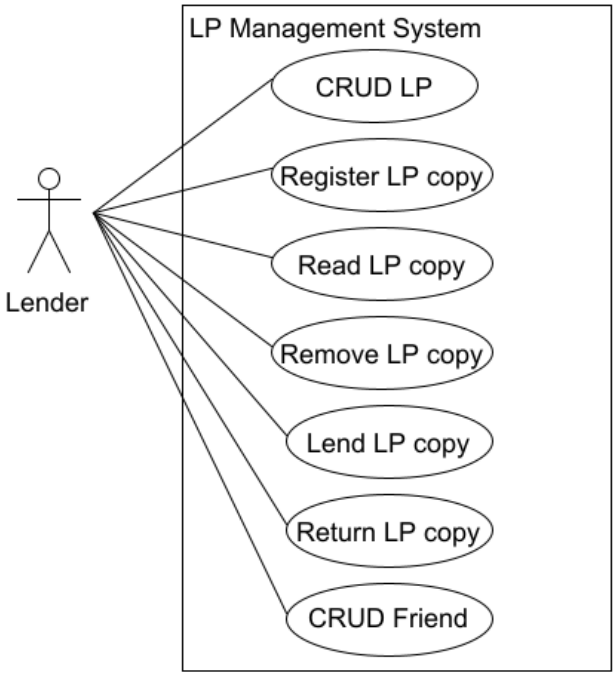
- **Friend Management**: Add, search, update, and delete friend profiles.
- **LP Management**: Add new LPs, search, update information, and delete LPs. You can also register multiple copies under the same title.
- **Loan Tracking**: Track which friends have borrowed specific LP copies, along with the loan date and return status. The system will allow you to register when an LP is returned.

When a friend wishes to borrow an LP, they can browse your collection and select a specific copy. You then register the loan in the system, recording both the friend and the chosen LP copy.

Use Case Model

Use Cases for the LP management system.

The following use cases are identified in the system:



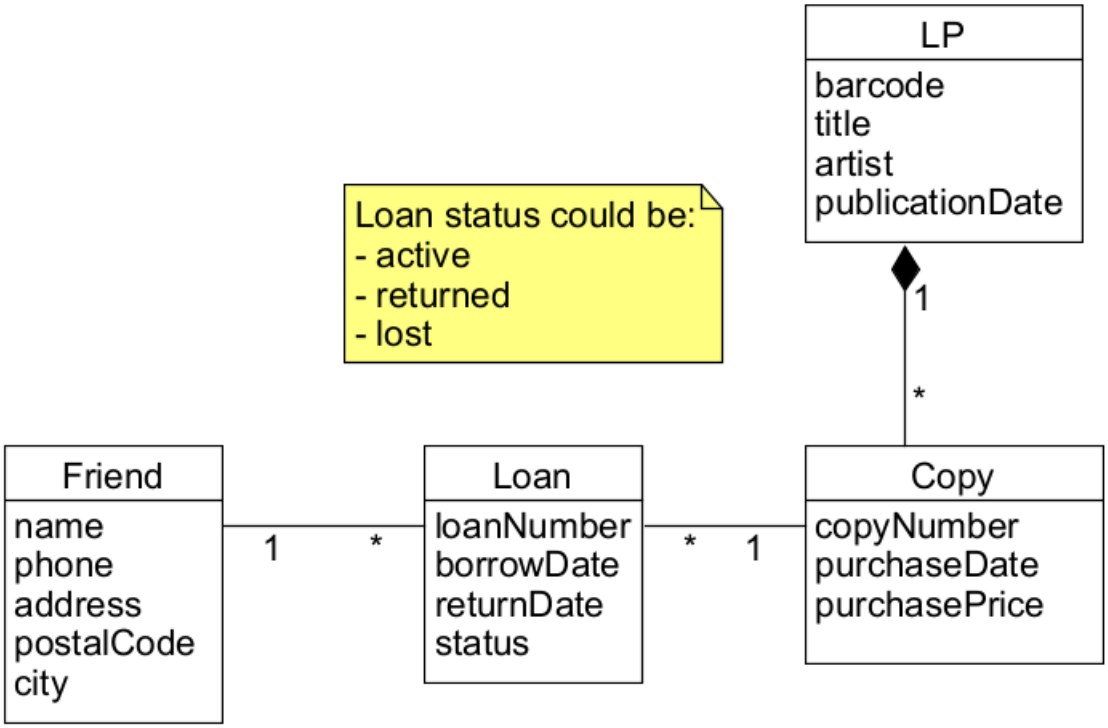
Use Case Diagram for LP Management System™

Prioritized use cases:

Use case name	*Business Value*	*Technical risk*	*Coverage*	**Total Score**	**Priority**	*UP phase*
Lend LP Copy	10	8	10	800	1	Elaboration
Register LP Copy	8	7	5	280	2	Construction
CRUD LP	8	5	1	40	3	Construction
CRUD Friend	7	5	1	35	4	Construction
Return LP Copy	4	4	2	32	5	Construction
Read LP Copy	3	4	2	24	6	Construction
Remove LP Copy	2	5	2	20	7	Construction

Domain Model for LP lending

Based on the use case descriptions, the following domain model has been developed:



Domain Model for LP Management System®.

Working on use cases

You work on the use cases one by one, starting with the highest priority. For each use case, you complete the analysis, design, implementation and testing before moving on to the next in order of priority. In parallel with working on the use case, it is a good idea to also make progress on the report.

1st priority (Elaboration phase)

Use case: "Lend LP Copy"

In this priority the most complex and highest prioritized use case "Lend LP Copy" must be designed, implemented, and tested according to the UP method. You will base it on the Domain Model already created.

First draft of fully dressed description of "Lend LP Copy":

Pre: Friend and Copy exist.

Post: A Loan is created and associated with Friend and Copy if the wanted copy is present.

1. A friend wants to borrow an LP.
2. The lender types in name or phone number.
3. The system finds the friend.
4. The lender states which copy there is to be borrowed.
5. The system returns copy-information.
6. The lender completes the loan.
7. The system records the copy and reports that the loan has been created.

Work for the 1st priority use case:

1. Complete the fully dressed use case by adding extensions (alternative flows).
2. Create system sequence diagram, and operation contracts.
3. Create the communication diagram. When creating the communication diagram, keep in mind that you are working with a layered architecture. Use this architectural structure, along with the domain model and operation contracts, to identify the objects and the methods to include in the communication diagram.
4. In parallel with the communication diagram, draw the design class diagram.
5. Describe your code standard. Extending it as you go along. *For inspiration:*
   a. *Java Coding Conventions : http://www.oracle.com/technetwork/java/codeconventions-150003.pdf*
   b. *[BlueJ] Appendix J* (Note does not completely follow the official conventions from 5.a.)
6. Setup your git repository in GitHub (see the document **Collaborating on Code with GitHub and GitHub Desktop)**
7. Write the code for the use case based on the communication diagram and the design class diagram.
8. Create a MainMenuUI class in the tui layer. The main menu should provide access to all other menus, and submenus should allow navigation back to the main menu. You decide on the design of the all the menus. If you need inspiration, refer to the example below, the SRW-skeleton from System Development, or one of the two projects included in this mini-project. The Main Menu could look like this (for non-English speaking customers, you should localize the user interface to fit their preferred language).

```
****** Main menu ******
(1) Friends menu
(2) LP menu
(3) Loan menu
(9) Generate test data
(0) Quit the program

Choice:
```

9. Create a `TryMe` class in the TUI layer which will populate `FriendContainer` and `LPContainer` with test data.
10. Implement tests for the use case.

2nd priority (Construction phase)

Use case: "Register LP Copy"
The "Register LP Copy" use case must be designed, implemented, and tested.

Casual description:
When a lender acquires a new copy of an LP, they must register it in the system. If the LP does not exist in the system, the use case terminates, and the lender must first create the LP. Otherwise, if the LP already exists, the lender finds the LP in the system and registers the new copy.

Work for the 2nd priority use case:

1. Create a fully dressed use case description, system sequence diagram and operation contracts.
2. Create the interaction diagram. Refer to the design class diagram from the previous iteration to see if any methods can be reused.
3. Update the design class diagram as needed.
4. Implement the use case based on the updated design.
5. Implement and conduct tests to ensure functionality and correctness.

3rd priority (Construction phase)

Use case: "CRUD LP"
The "CRUD LP" use cases must be designed, implemented, and tested.

Work for the 3rd priority use case:

1. Create the relevant artefacts (SSD, operation contracts and communication diagrams) for the four use cases in the CRUD LP.
2. Update the design class diagram.
3. Implement the use case.
4. Implement and conduct tests.

4th priority (Construction phase)

Use case: "CRUD Friend"
The "CRUD Friend" use cases must be designed, implemented, and tested.

Work for 4th priority:

1. Create the relevant artefacts (SSD, operation contracts and communication diagrams) for the four use cases in the CRUD Friend.
2. Update the design class diagram.
3. Implement the use case.
4. Implement and conduct tests.

5th priority (Construction phase)

Use case: "Return LP copy"
In this priority the use case "Return LP copy" must be designed and implemented

Casual description:
A friend wants to return an LP copy.
If the LP copy is valid and loaned, the system records the return of the LP copy.
If the LP copy is not currently loaned, the system displays a message notifying that the LP copy was not on loan.
If the LP copy does not exist, the system displays an error message indicating that the LP copy is not found.

Work for 5th priority:

1. Create fully dressed description, SSD, operation contracts and communication diagrams for the use case.
2. Update the design class diagram.
3. Implement the use case.
4. Implement and conduct tests.

Remember to check the documentation requirements when you are writing your report!